

Removal of Noise from Electrocardiogram Using Digital FIR and IIR Filters with Various Methods

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Abstract-- Electrocardiogram (ECG) is a type of measuring the electrical activities of heart. Each section of ECG is necessary for the diagnosis of various cardiac problems. But the amplitude and time period of ECG signal is generally corrupted by various noises. After an analog ECG signal is transformed into digital format, appropriate digital filter can be utilized to repress the various kinds of noise like Baseline Wander, Power line Interference, High –frequency Noise, Physiological Artifacts etc., depends on their specifications. In generic two types of method can be classified in this paper; FIR filters like Rectangular, Hann, Blackman, Hamming and Kaiser window techniques and IIR filters like Butterworth, Chebyshev I, Chebyshev II and Elliptic filters are also prospected to reduce artifacts in ECG signal. The results are collected from different orders for FIR filter as 56, 300, 450, and 600 and for IIR filter as 1, 2, and 3. The signals taken from the MIT-BIH data base which contains the normal and abnormal waveforms. The work has been implemented in MATLAB FDA Tool. The results are obtained using different window based FIR filters, IIR filter with different approximation methods and their respective waveforms are shown. In addition, power spectrum density, signal to noise ratio (SNR) and means square error (MSE) of both noisy and filtered ECG signals are calculated. We observed that Digital FIR filter with Kaiser Window in order 56 shows high performance as compared to the other windowing techniques and Digital IIR filter approximation methods.

Index Terms—FIR filters; IIR filter; Electrocardiogram; SNR; MSE; MATLAB

I.INTRODUCTION

Signal processing has a wealthy background and its significance in biomedical engineering is known to all. An electrocardiogram demonstrates the electrical activity in the heart, and may be analyzed in characteristic parts, named P, Q, R, S, and T waves. When an ECG is recorded, it would be corrupted with various kinds of noise. Therefore exploitation of pure ECG indices from noisy measurements has been one of the important considerations of biomedical signal processing and requires trusty techniques to maintain the diagnostic data of the recorded signal. An ECG is very sensitive, different

types of artifact and interference can contaminate the ECG signal as the real amplitude and time period of the signal can be changed. ECG signals are mostly affected by:

- 1) *Baseline Wander*: It is actually the effect where the base axis of any ECG signal viewed on a screen appears to “wander” or move up and down rather than be straight. This cause the entire signal to shift from its normal base. It is usually caused due to improper electrodes and by respiration or movement of the patient.
- 2) *Physiological Artifacts*: This kind of noises is primarily produced by other organs of the body or caused by muscular retractions relevant to breathing. Electrode movement artifact is commonly considered to be the most difficult, since it may imitate the figure of abnormal beats and cannot be eliminated simply by normal filters, as can an artifact of other kinds.
- 3) *High –frequency Noise*: This kind of random artifact could be due to the thermal cause in electrodes, the recording system, , the instrumentation amplifiers and pickup of ambient electromagnetic signals by cables[1].In the real-time clinical monitoring systems in surgery, electrosurgical noise is a considerable barrier to be overcome.
- 4) *Power line Interference*: Interference caused by radiation from the high voltage power for Examples of environmental interference are those effected by 50 or 60 Hz power-supply lines, , radiation from lights, electrodes motion and radio-frequency emissions from nearby medical devices.

These artifacts and interference makes the wrong diagnosis of the ECG signal [1-2].So, the elimination of these artifacts and interference from ECG signal has become very difficult. Different kinds of digital filters have been used to rectify the problem [2-4].But it is hard

to utilize these filters with fixed coefficients to remove several kinds of artifacts, because the ECG signal is known as a non-stationary signals.

A number of researchers have worked on the digital filters for eliminating from ECG signal and various techniques have been offered or proposed. Chandrakar et al. studied FIR and IIR filters then concluded that Kaiser Window based FIR filters are better to remove artifacts from ECG signals [5]. They implemented efficient hardware architecture for high speed FIR filter for electromyogram removal from electrocardiogram signals [6]. FIR digital filters were designed with new window to remove the interferences or the artifacts and calculated SNR of noiseless ECG compared with existing windows [7]. They proposed second order infinite impulse response (IIR) notch filter, adaptive notch filtering technique with least mean square algorithm and discrete wavelet transform method for the removal of power line interference from ECG signal and considered two synthesis parameters MSE and SNR [8]. Mohandas Choudhary, Ravindra Pratap Narwaria "Suppression of Noise in ECG Signal Using Low pass IIR Filters" describes an IIR filter design for noise removing [9] it gives various types of IIR filter design but the result of Butterworth is best among all. Mbachu et al. designed digital FIR low pass, high pass and notch filters with a rectangular window for the removal of the artifacts [10]. In this work, the artifact from the electrocardiogram waveform has been eliminated utilizing these various methods. Signal to noise ratio (SNR), Means square error (MSE) and power spectral density resulting from the algorithms has been investigated and compared.

The rest of this article organized as follows: Section II theoretical description of digital filters. Section III summarizes the techniques investigated. Section IV provides the results and output waveforms. Finally, some conclusions are demonstrated in Section V.

II. THEORY

A) Digital IIR filter

Digital Filters are designed by using the values of both the past outputs and the present input, an action brought about by convolution. If such a filter is subjected to an impulse then its output not requirement essentially become zero. The impulse response of such a filter may be infinite in period. Such a filter is known as an Infinite Impulse Response filter or IIR filter. The infinite impulse response of such a filter implies the capability of the filter to have an infinite impulse response. This indicates that the system is prone to feedback and inconstancy. IIR filters are designed fundamentally by the Impulse inmutability or the Bilinear Transformation method. The IIR filter is determined by the equation below:

$$y[n] = \sum_{k=0}^{N-1} h(k) \cdot X(n-k) \quad (1)$$

B) Digital FIR filter

FIR filters are digital filters with finite impulse response. They are also known as non-recursive digital filters as they do not have the feedback, even though recursive algorithms can be used for FIR filter realization. FIR filter is characterized by following equations:

$$y(m) = \sum_{k=1}^N a_k y(m-k) + \sum_{k=0}^M b_k x(m-k) \quad (2)$$

In the window design method, one first designs an ideal IIR filter and then truncates the infinite impulse response by multiplying it with a finite length window function. The result is a finite impulse response filter whose frequency response is modified from that of the IIR filter. Multiplying the infinite impulse by the window function in the time domain results in the frequency response of the IIR being convolved with the Fourier transform of the window function. If the window's main lobe is narrow, the composite frequency response remains close to that of the ideal IIR filter.

III. METHODOLOGY

ECG signal fundamentally contains frequency between 0-250Hz. Research proves that the frequency range of the ECG signal is 0-250 Hz, The sampling frequency was selected to facilitate performances of 60 Hz digital notch filter in arrhythmia detectors, sampling frequency of data signal is 360 Hz and amplitude 1mv. We designed the filter for corrupted ECG signal in four steps: In the first, step with the help of FDA Tool in MATLAB software design digital FIR filters with various windowing techniques like: Rectangular, Hann, Hamming, Blackman and Kaiser with high pass filter cut off frequency 0.5 Hz were utilized to delete baseline wander noise from noisy ECG signal. In the second step, removing power line interference (50/60 Hz) by band stop with cut off frequency (59.5Hz-60.5 Hz) was carried out, in the third step, we deleted EMG noise by applying low pass filter with cut off frequency 100Hz, in order 56,300,450 and 600, finally moving average filter to smooth the ECG waveform. The task was accomplished in various orders.



Fig1: Block diagram of ECG signal processor

In addition, we designed IIR digital filter with different approximation methods like Butterworth, Chebyshev I, Chebyshev II, and Elliptic in order 1, 2 and 3 with the low pass, high pass, and band stop filters the specifications mentioned above. The efficiency analysis contained the comparison of outcomes generated by filters designed during the modelling method by replacing

filter parameters arranging them so that the one where outcomes obtained were best.

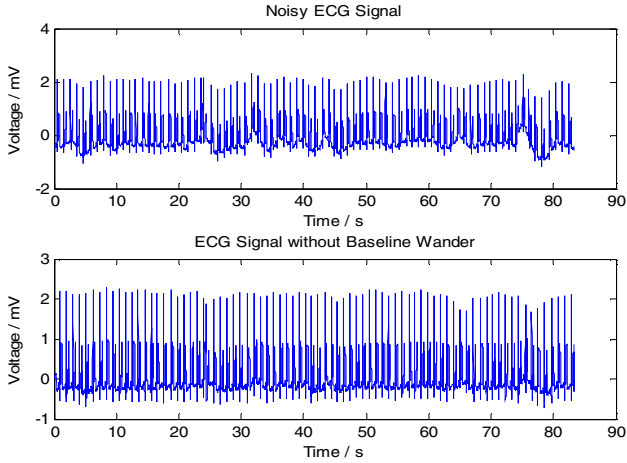


Fig2: Noisy ECG (113m) and removed BLW using Kaiser Window

The results were collected with the performing filters through the ECG database from MIT-BIH site. The ECG samples 113m, 114m, 116m, 118m, 124m, 201m, 203m, 205m (MLII, VI) obtained from MIT have supported then own research into arrhythmia database which consists 48 half hours excerpts of two channel ambulatory ECG recordings utilized to verify the results of digital filter designed as described in the above methodology.

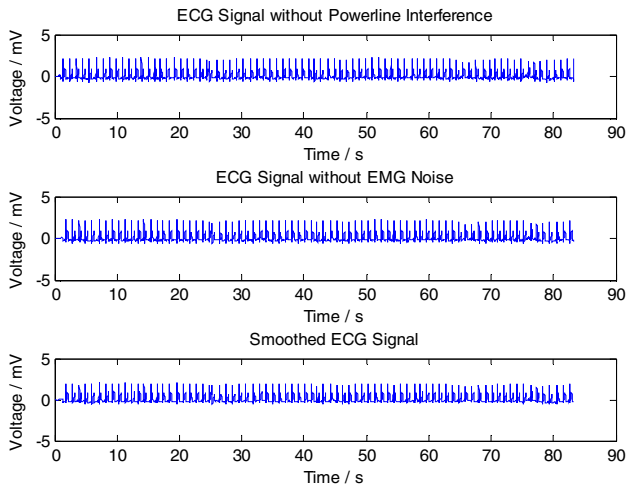


Fig3: Remove PLI (60 Hz) and EMG using Kaiser Window

IV. RESULTS

The results were generated with the designed FIR, IIR filters applied to various ECG signals from MIT-BIH database. The filters like high pass, band stop and low pass with various windowing at order 56, 300, 450, 600 and IIR with different approximation methods in order 1, 2 and 3 shows different results. The graphs for the waveforms and their power spectral density before and after filtering are shown for various windows and IIR filters.

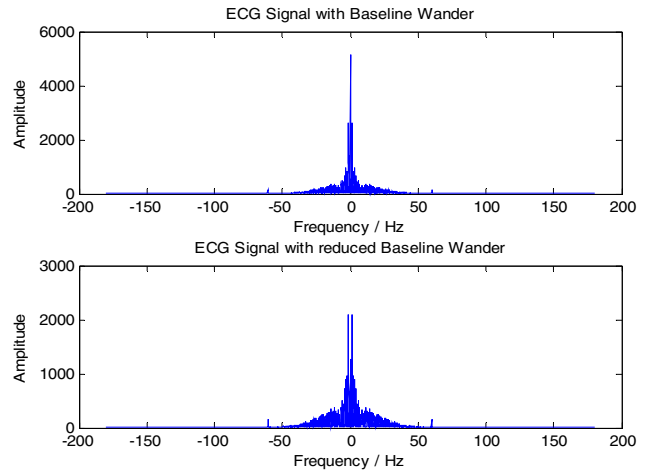


Fig4: Frequency spectrum of with and without BLW using Kaiser Window

By using these windows and IIR approximation methods, we designed the high pass filter of cut-off frequency 0.5 Hz for removing baseline wandering, band stop filter of cut-off frequency 59.50Hz – 60.5 Hz for removing power line interference and low pass filter of cut-off frequency 100Hz for removing EMG noise. The comparison of different FIR and IIR filters by calculation of SNR (Signal to noise ratio) and MSE (mean square error) was done at 56 and 1 orders. The results are shown in tabular form:

Table 1: SNR comparison of window based FIR filters

Real ECG data	SNR of noisy ECG	Signal to noise ratio of FIR filtered ECG signal				
		Rectangular	Hann	Hamming	Blackman	Kaiser
113	5.77	6.72	6.13	6.17	6.05	6.69
114	1.36	2.52	1.82	1.87	1.72	2.49
116	8.00	9.13	9.02	9.03	8.91	9.16
118	8.96	10.02	9.29	9.35	9.20	10.05
124	8.62	9.86	9.03	9.09	8.92	9.89
201	14.37	14.47	14.53	14.53	14.51	14.48
203	12.84	13.71	13.08	13.13	13.01	13.76
205	9.02	9.94	9.50	9.54	9.41	9.95

Table 2: MSE comparison of window based FIR filters

Real ECG data	SNR of noisy ECG	Mean square error of the FIR filtered ECG signal				
		Rectangular	Hann	Hamming	Blackman	Kaiser
113	0.21	0.17	0.19	0.19	0.18	0.17
114	0.03	0.02	0.03	0.03	0.03	0.01
116	1.26	0.98	0.10	1.09	1.12	0.97
118	1.09	0.81	0.94	0.93	0.96	0.80
124	0.76	0.56	0.65	0.64	0.67	0.55
201	0.06	0.05	0.62	0.06	0.06	0.05
203	0.29	0.24	0.25	0.25	0.02	0.23
205	0.15	0.11	0.13	0.13	0.13	0.10

Table 3: SNR comparison of IIR approximation method

Real ECG data	SNR of noisy ECG	Signal to noise ratio of the IIR filtered ECG signal			
		Butterworth	Chebyshev Type 1	Chebyshev Type 2	Elliptic
113	5.77	5.70	5.37	1.61	5.37
114	1.36	1.94	1.39	2.62	1.39
116	8.00	8.40	8.52	0.94	8.52
118	8.96	8.64	8.42	10.35	8.42
124	8.62	9.46	8.60	0.74	8.60
201	14.37	14.28	14.33	8.16	14.33
203	12.84	12.31	12.61	3.10	12.61
205	9.02	9.02	9.01	5.47	9.01

Table 4: MSE comparison of IIR approximation method

Real ECG data	MSE Of noisy ECG	MSE of IIR filtered ECG signal			
		Butterworth	Chebyshev Type 1	Chebyshev Type 2	Elliptic
113	0.21	0.14	0.15	2.59	0.15
114	0.03	0.01	0.11	3.48	0.01
116	1.26	0.34	0.35	5.55	0.35
118	1.09	0.15	0.16	6.27	0.16
124	0.76	0.11	0.12	4.20	0.12
201	0.06	0.14	0.14	2.56	0.03
203	0.29	0.03	0.03	3.87	0.21
205	0.15	0.19	0.21	6.80	0.03

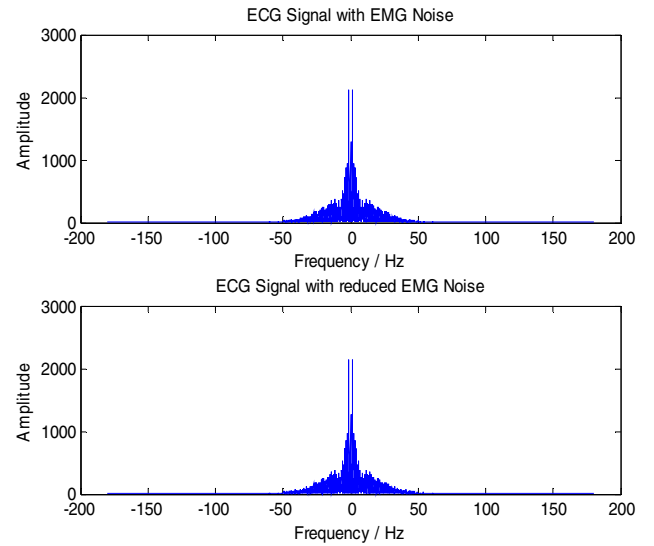


Fig6: Frequency spectrum of with and without EMG using Kaiser Window

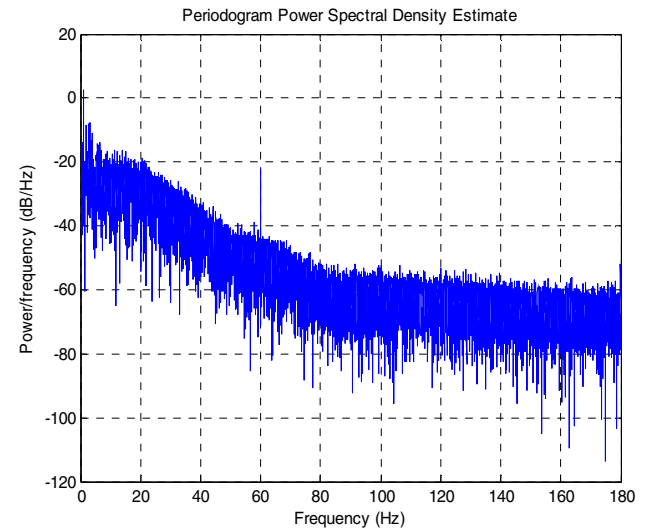


Fig7: Power spectrum density of noisy ECG signal using Kaiser Window

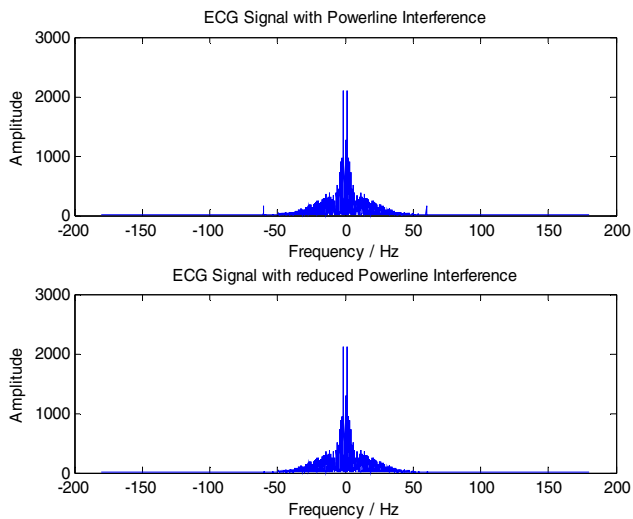


Fig5: Frequency spectrum of with and without PLI using Kaiser Window

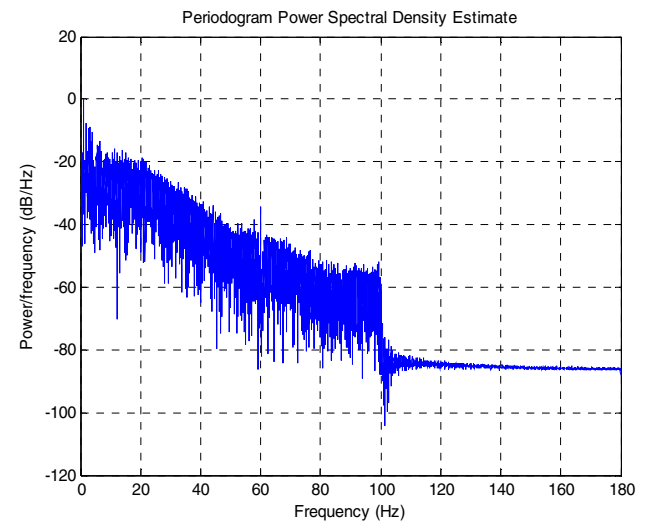


Fig8: Power spectrum density of denoised ECG signal using Kaiser Window

V. CONCLUSION

This article, introduced a method used for artifacts reduction from ECG signal which basically includes designing of filter with different windows like Rectangular, Hann, Hamming, Blackman and Kaiser Window with different order 56,300,450,600 and IIR filters approximation methods like Butterworth, Chebyshev I, Chebyshev II and Elliptic at orders 1, 2, 3 with proper parameters indicating the best outcomes of baseline wander noise, power line interference and electromyography noise removal. The results for various filters are considered and evaluated by waveforms, power spectrums density (PSD), signal to noise ratio (SNR) and Mean square error (MSE).The Kaiser window order 56 of FIR filter showing the best results was compared to all orders in FIR as well as IIR filters at order 1, 2 and 3.

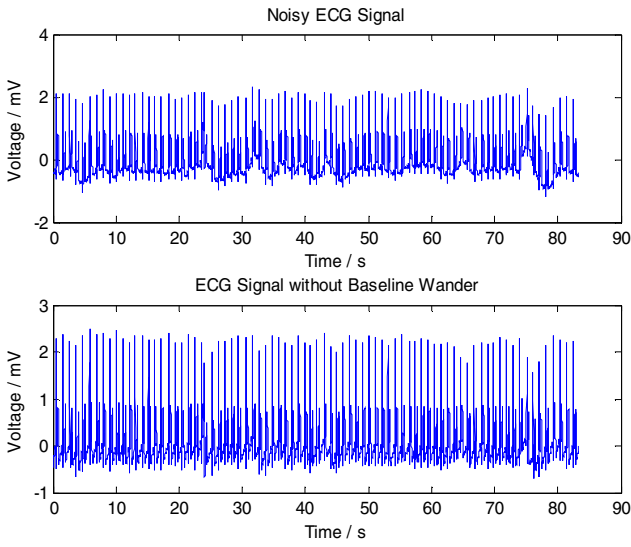


Fig9: Noisy ECG (113m) and removed BLW using Butterworth approximation method

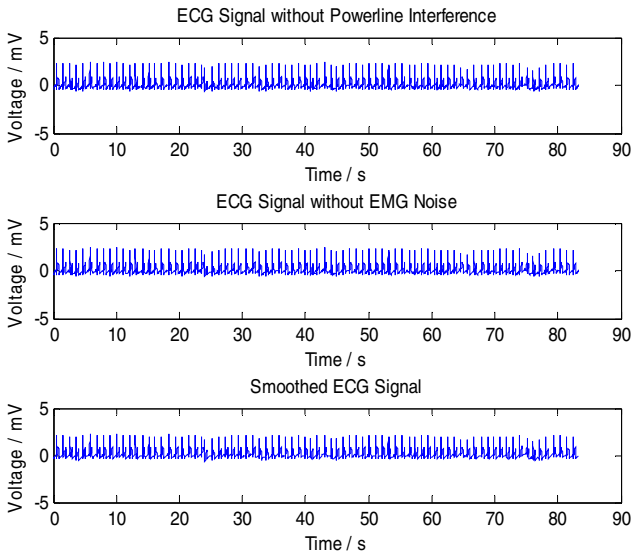


Fig10: Remove PLI (60 Hz) and EMG using Butterworth approximation method

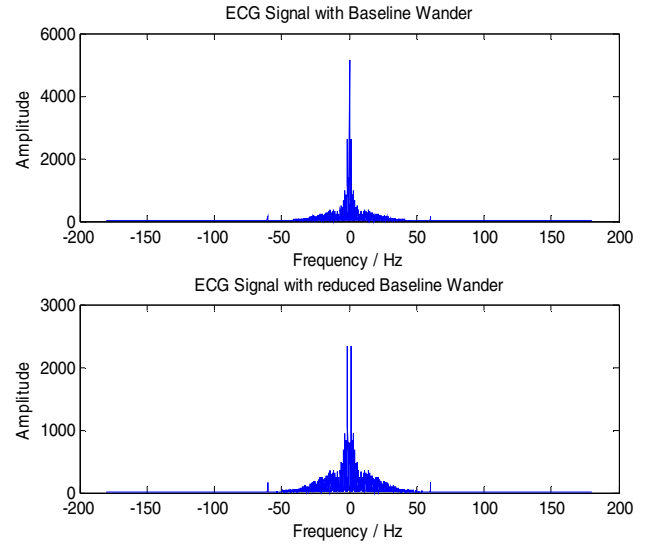


Fig11: Frequency spectrum of with and without BLW using Butterworth approximation method

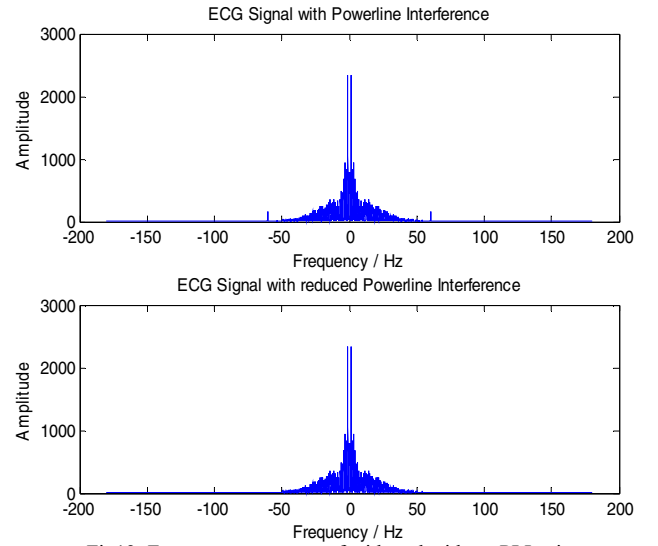


Fig12: Frequency spectrum of with and without PLI using Butterworth approximation method

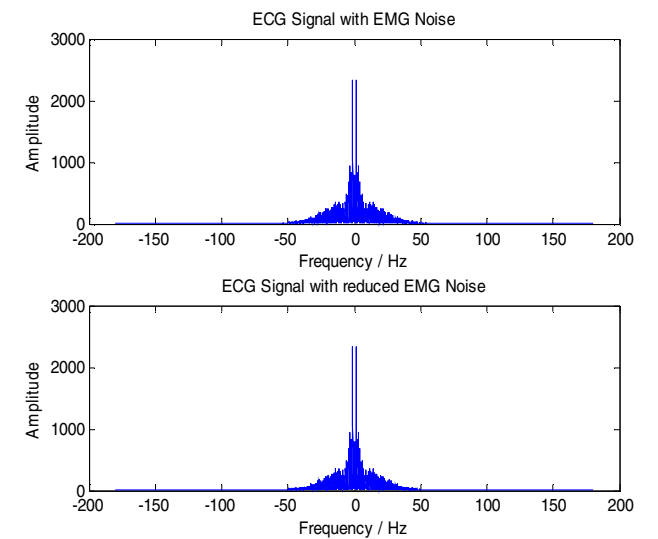


Fig13: Frequency spectrum of with and without EMG using Butterworth approximation method

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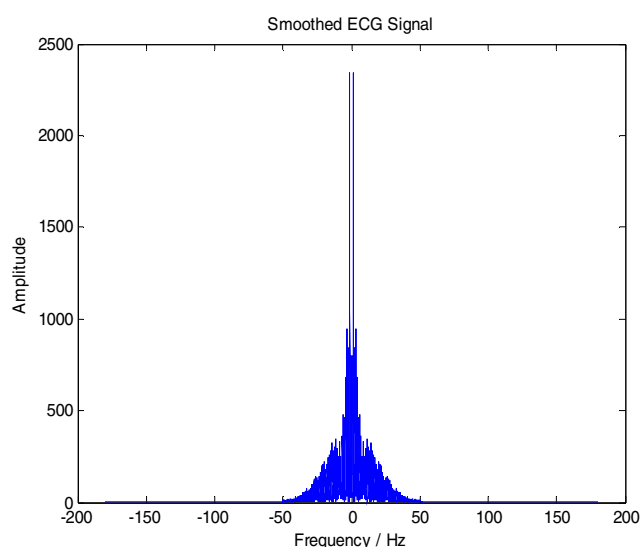


Fig.14: Smoothed ECG (113) Signal (FFT) for Butterworth approximation method

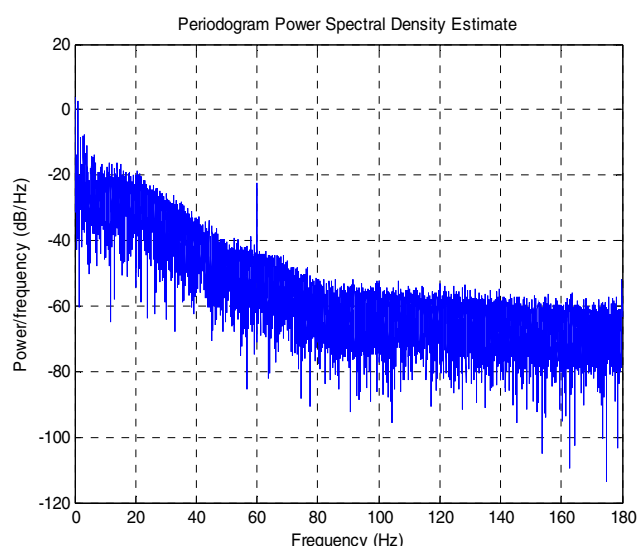


Fig15: Power spectrum density of noisy ECG signal using Butterworth approximation method

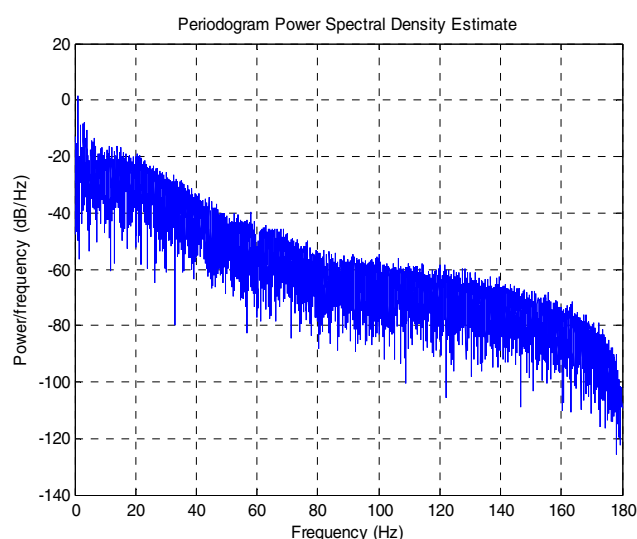


Fig16: Power spectrum density of denoised ECG signal using Butterworth approximation method